

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

Learn about the mysterious Nazca lines and what they could mean in Earth. Then in Ignorance, we try to figure how planes fly, after that take a closer look at ISRO's SSLV in Space Age and more...



Galactic mystery solved

With the help of thousands of volunteers, scientists have managed to figure out the source of mysterious gamma rays that has been puzzling researchers for more than 20 years. The object has been identified as a neutron star with a very low mass companion in orbit around it.

<http://dgit.in/caseclosed>

WHAT'S NEW

Using microplastics to remove microplastics

Scientists have come up with a self-sustaining system that addresses the global problem of microplastic pollution in the oceans, that has reached more places and in more quantity than no one had ever anticipated. Since it is all but impossible to filter out the ocean, the problem has to be tackled using unique approaches. The approach uses sticky plastic microparticles, that are binded to peptides and so called soft dendritic materials, to create a circular system in the ocean. These microcleaning particles would capture the microplastics from the oceans,



collapse into themselves, and form clumps that float up to the surface of the ocean, from where

they can be easily removed. Both the micro-cleaning plastics and the polluting microplastics would then be processed by a bioreactor, that breaks down the microplastics to create more microcleaning plastics, as well as other potentially useful materials.

<http://dgit.in/rmvplstc>

The Milky Way collided with another galaxy 3 billion years ago

When a galaxy collides with another, the stars of the smaller galaxy are pulled into a long cord of stars that move in unison in a long, sharp curve around the halo of the galaxy. Such formations are known as shell structures and have been identified in other galaxies. Astronomers had long suspected that such shell structures exist within the Milky Way too.



Disk galaxies such as the Milky Way have fewer shell structures than spherical galaxies, which have more prominent and well-understood shells. Thanks to the a high resolution map of all the

stars in the galaxy obtained by the Gaia instrument, researchers have been able to identify shell structures in our own galaxy, which point to a collision with a dwarf galaxy about 3 billion years ago. The researchers were able to identify four shell structures in the galaxy, two in the Virgo overdensity, and two in the Hercules Aquila cloud region.

<http://dgit.in/glxysshells>



Genomic Rosetta stone

Humans and chimps share 90 percent of the DNA, but are very different because of how the genes are used and regulated. Researchers have found a new tool that allows them to better understand this regulation, which paves the way for further insights into how DNA works.

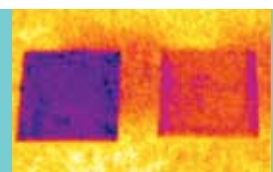
<http://dgit.in/genosetta>



Monkeys help AI

MIT scientists are studying the brains of living monkeys to test how back and forth information processing in neural networks can improve object recognition. They studied a neural pathway in monkeys that allows the primates to rapidly differentiate between a bird and an elephant.

<http://dgit.in/monkai>



Paint to replace ACs

Researchers from Purdue University have come up with a paint that can replace air conditioners. The white paint reflects the sunlight away from the house and prevents it from getting warm in the first place. The researchers are now developing more shades.

<http://dgit.in/paintac>